LAB ASSIGNMENT 9

Landing AI: Vision-Based Cognitive System

Cognitive Computing (UCS420)

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1. Objective

To build, train, test and deploy an object detection model on the LandingLens platform. The model is trained to locate the label on a medicine ampoule from a small set of labelled images.

2. Dataset Preparation and Annotation

A set of 15 images of the ampoule was uploaded to the project. The label region in each image was marked manually with a bounding box under a single class, "Box". The images were split into training and dev sets.

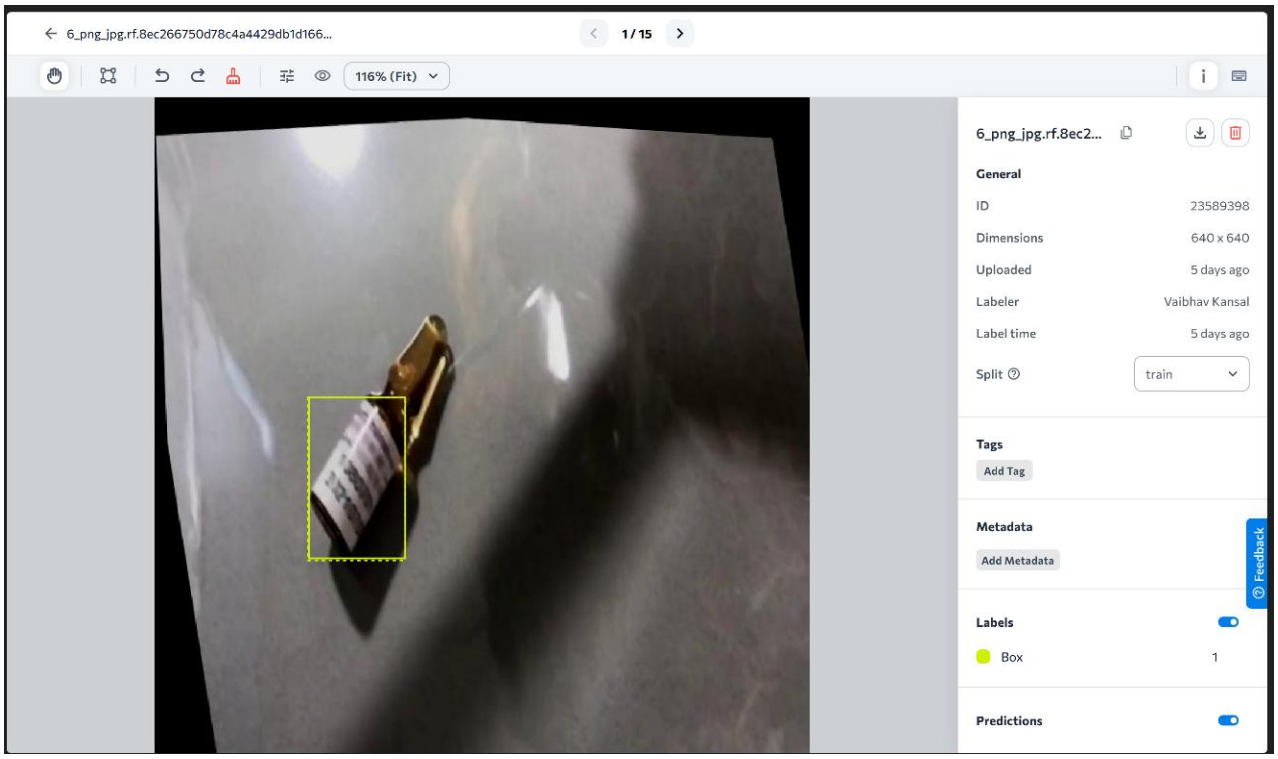


Figure 1: Bounding box drawn on the ampoule label (image 1 of 15)

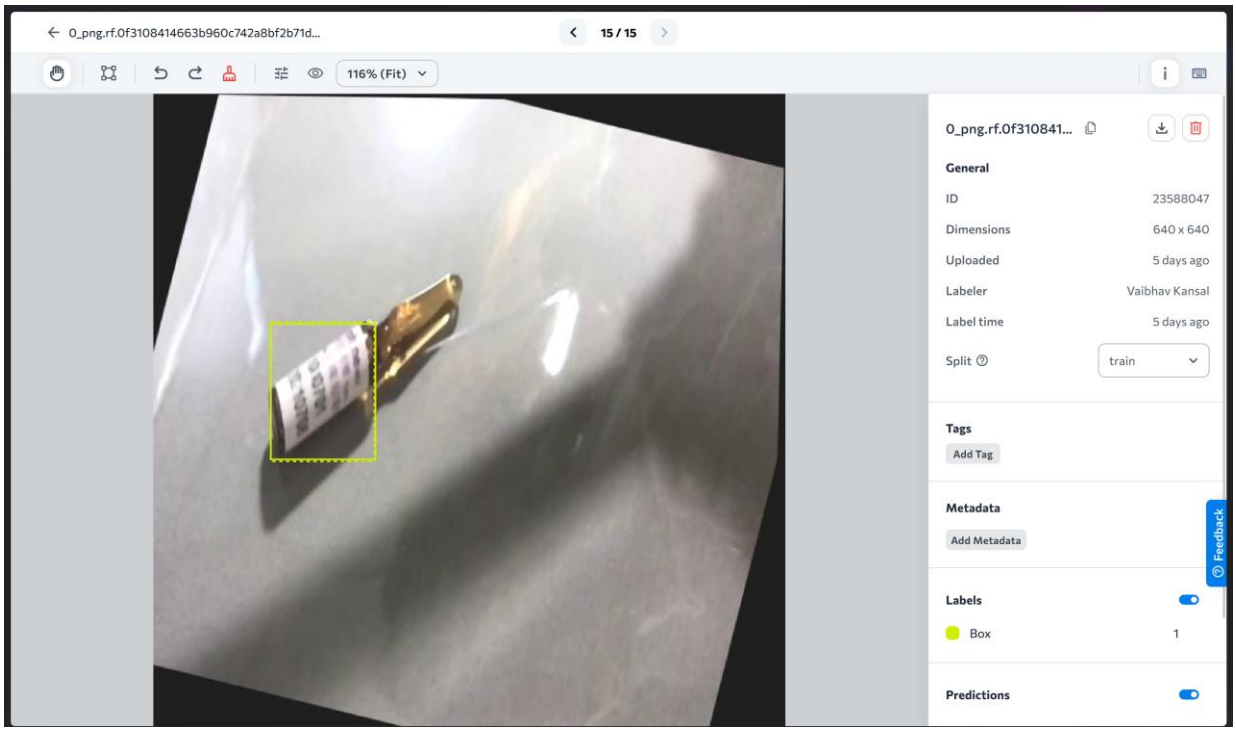


Figure 2: Annotation of the last image in the set (image 15 of 15)

3. Training

The model was trained from the Build page with 12 images in the training set and 3 in the dev set. The trained model is listed on the right of the screen as Model-10-27-2026_1.

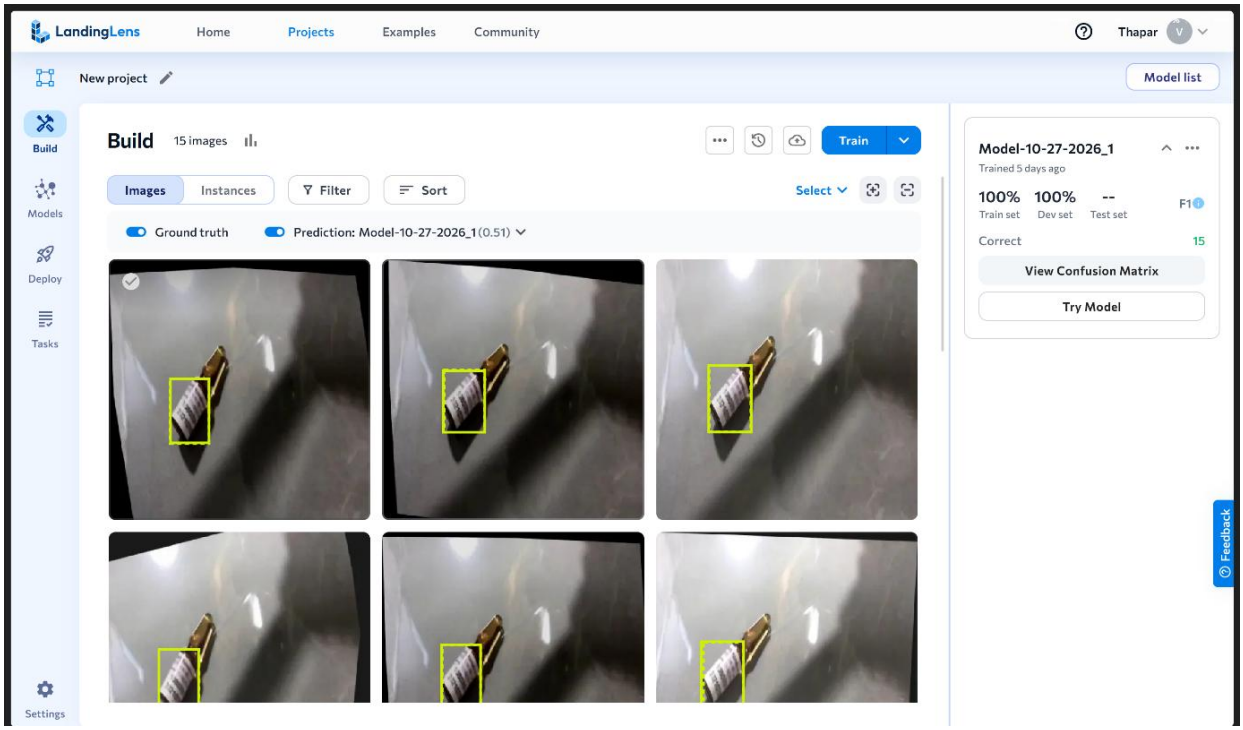


Figure 3: Labelled dataset and trained model on the Build page

4. Model Evaluation

The confusion matrix shows all 15 labelled boxes were detected correctly. Precision and recall are both 100%, and F1 is 100% on the training and dev sets. No test set was held out, so the test score is not available. The confidence threshold was 0.51.

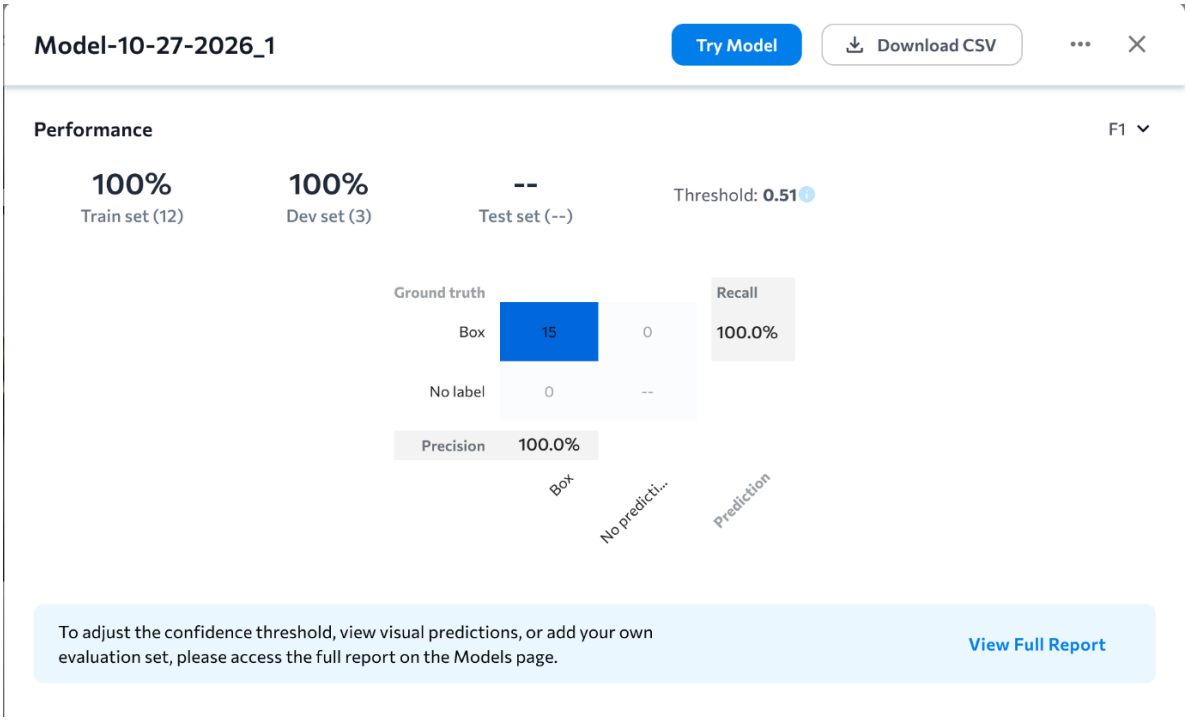


Figure 4: Performance summary and confusion matrix

5. Testing

The model was tried on two new images using a confidence threshold of 0.51. In both cases it detected the label and returned one "Box" prediction.

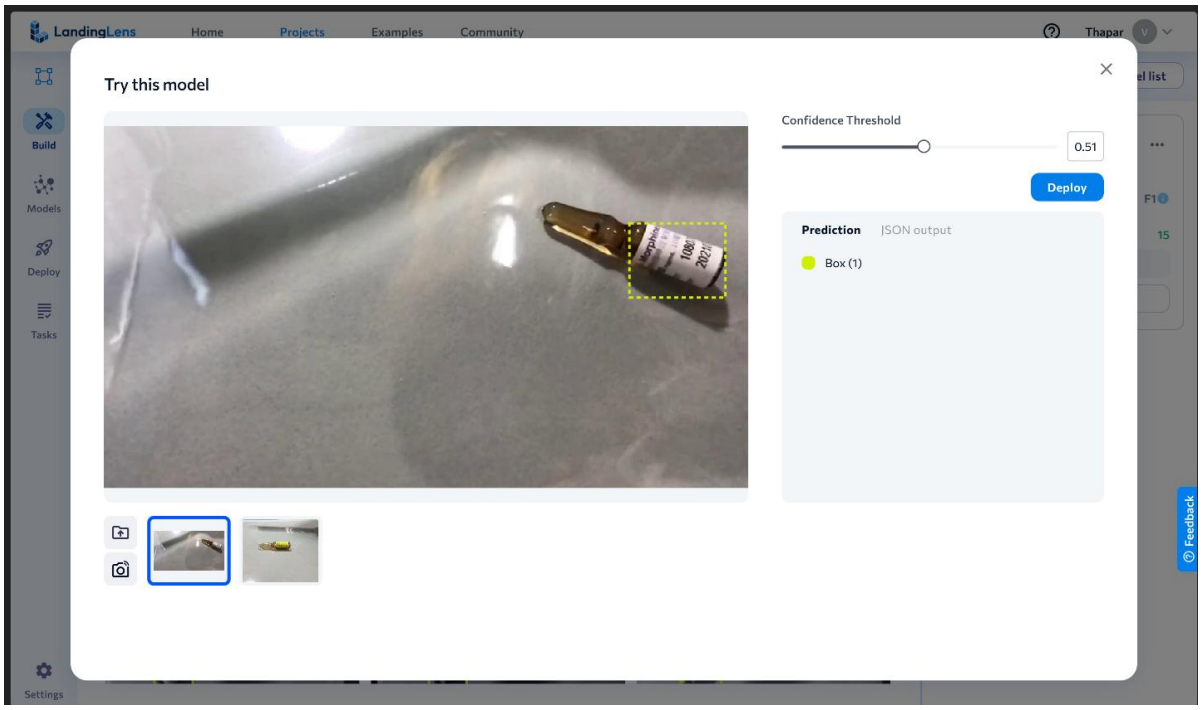


Figure 5: Test image 1, label detected

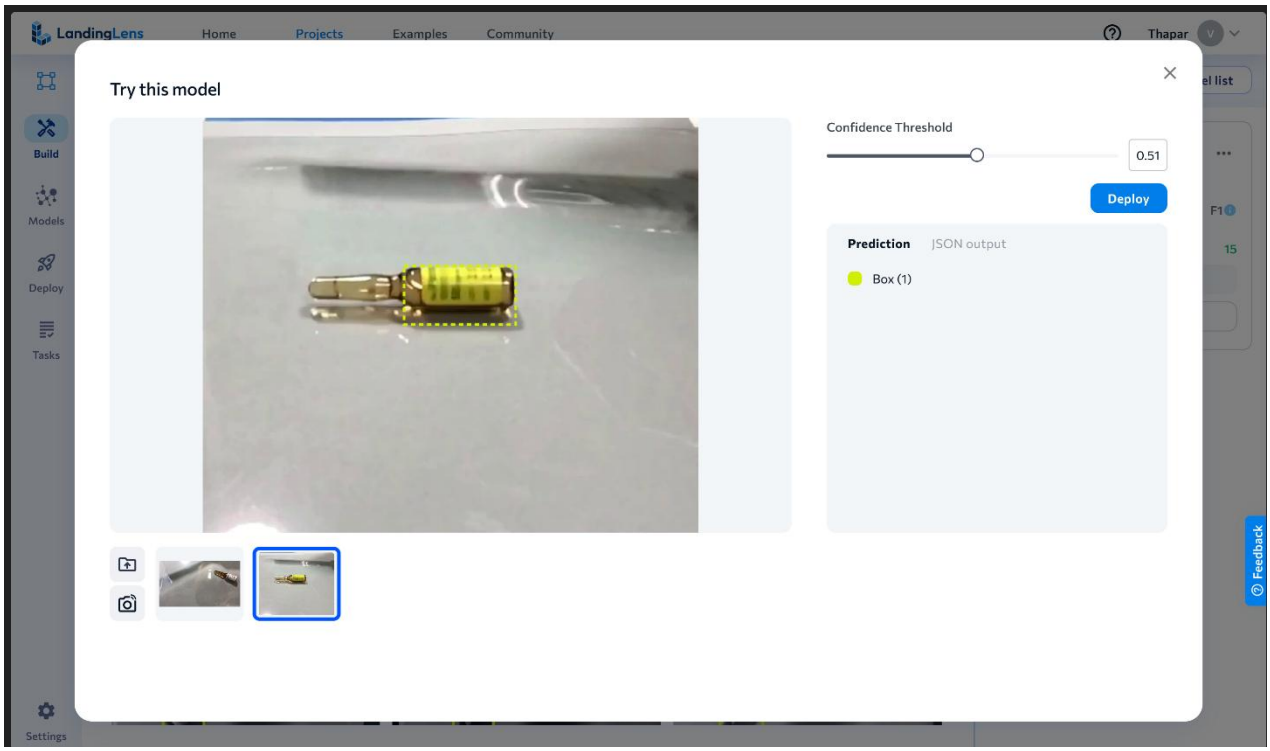


Figure 6: Test image 2, label detected

6. Deployment

The model was deployed as a cloud endpoint named medicine_label at the same 0.51 threshold. The Deploy page confirms the deployment and provides ready-made code for API inference (Python SDK, JavaScript SDK and cURL) and a QR code for mobile inference.

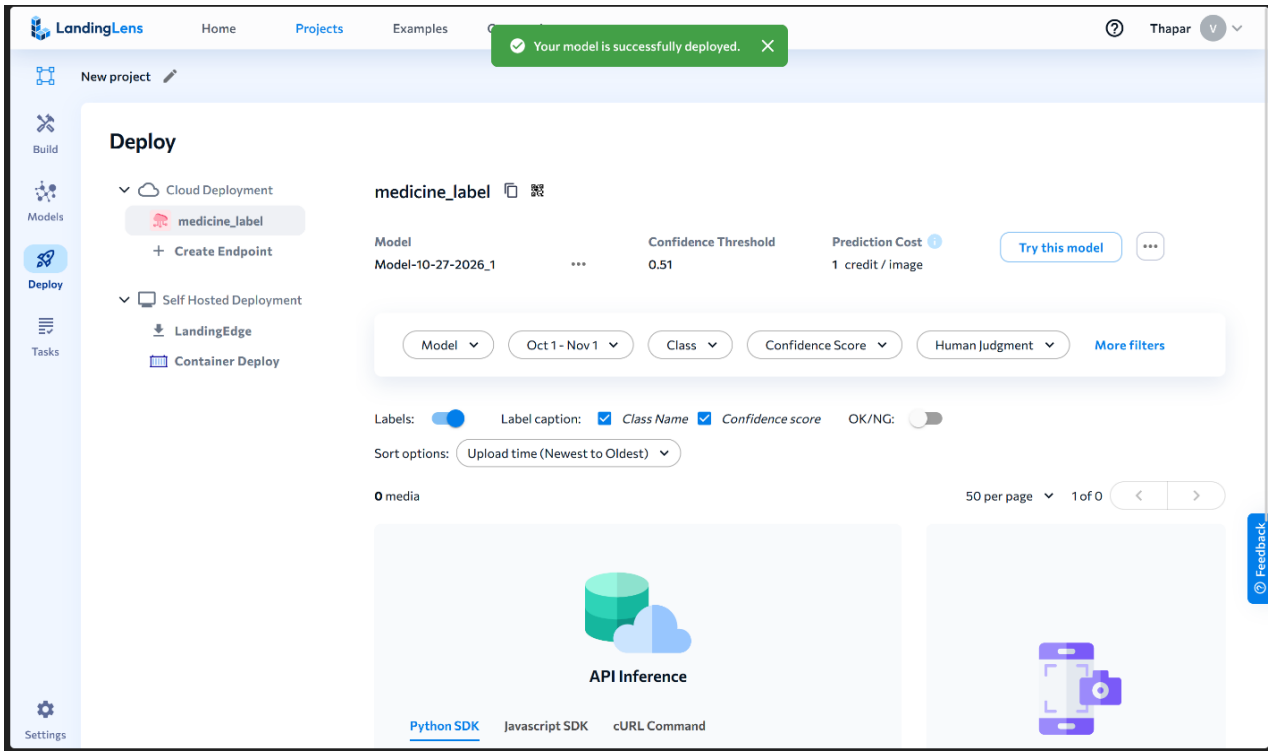


Figure 7: Successful deployment of the cloud endpoint

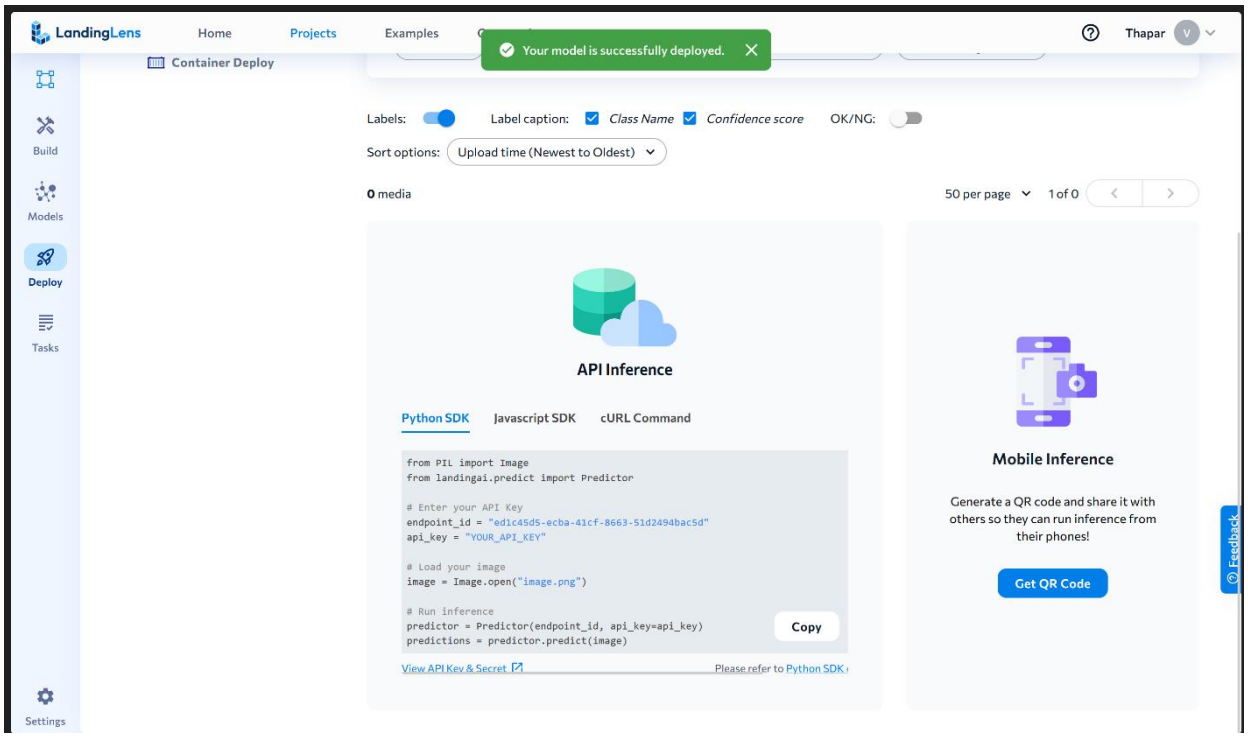


Figure 8: API inference options for the deployed model

7. Conclusion

The model detected the medicine label accurately on both the labelled data and the new test images, and was deployed successfully. Since the dataset has only 15 images and no separate test set, the 100% scores should be treated as indicative. A larger and more varied dataset would give a more reliable measure.

8. Python Code for API Inference

The code below was generated on the Deploy page for calling the deployed endpoint. The API key is a placeholder and has to be replaced with a personal key.

```
from PIL import Image
from landingai.predict import Predictor

# Enter your API Key
endpoint_id = "ed1c45d5-ecba-41cf-8663-51d2494bac5d"
api_key = "YOUR_API_KEY"

# Load your image
image = Image.open("image.png")

# Run inference
predictor = Predictor(endpoint_id, api_key=api_key)
predictions = predictor.predict(image)
```